

Project Design Phase-II

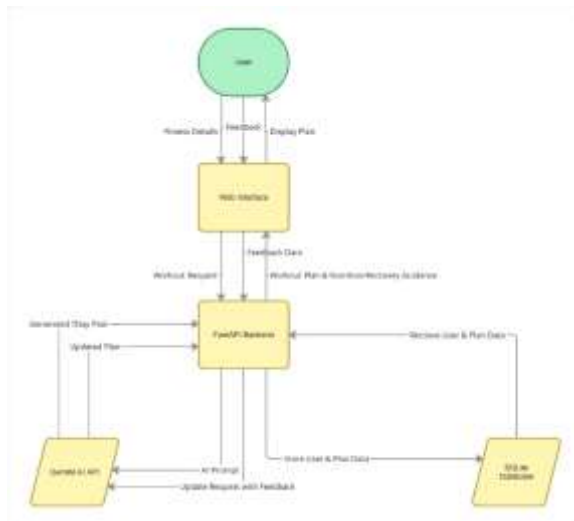
Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-3464
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Technical architecture diagram for a web application



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web interface through which users enter fitness details and view generated plans	HTML, CSS, JavaScript / Angular Js / React Js etc.
2.	Application Logic-1	Handles user inputs, API requests and application workflow	Python, FastAPI
3.	Application Logic-2	Generates personalized workout plans using AI	Gemini AI AP
4.	Application Logic-3	Processes user feedback and updates the workout plan	Python, FastAPI, Gemini AI
5.	Database	Stores user details, workout plans, feedback and history	SQLite
6.	Cloud Database	Cloud database service	Not used in current prototype
7.	File Storage	Stores application files and static resources	Other Storage Service or Local Filesystem
8.	External API-1	AI-based workout plan generation	Google Gemini API
9.	External API-2	No additional external API used	Not applicable.
10.	Machine Learning Model	Generates personalized fitness plans and updates based on feedback	Google Gemini Generative AI
11.	Infrastructure (Server / Cloud)	Application runs as a FastAPI web application	Local Server / Cloud Deployment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	FastAPI is used as the backend web framework. HTML, CSS and JavaScript are used for the web interface.	FastAPI, Python, HTML5, CSS3, JavaScript
2.	Security Implementations	User inputs are validated before processing. API credentials are protected and should not be exposed in the client-side interface.	Input Validation, API Key Protection
3.	Scalable Architecture	The application separates the user interface, backend application logic, AI service and database, allowing individual components to be enhanced independently	FastAPI, Gemini API, SQLite
4.	Availability	The application can be deployed on a web server so that users can access the FitBuddy interface through a browser.	FastAPI Server / Cloud Deployment
5.	Performance	The application sends only the required fitness information and feedback to the Gemini API and returns the generated workout plan to the user.	FastAPI, Gemini API